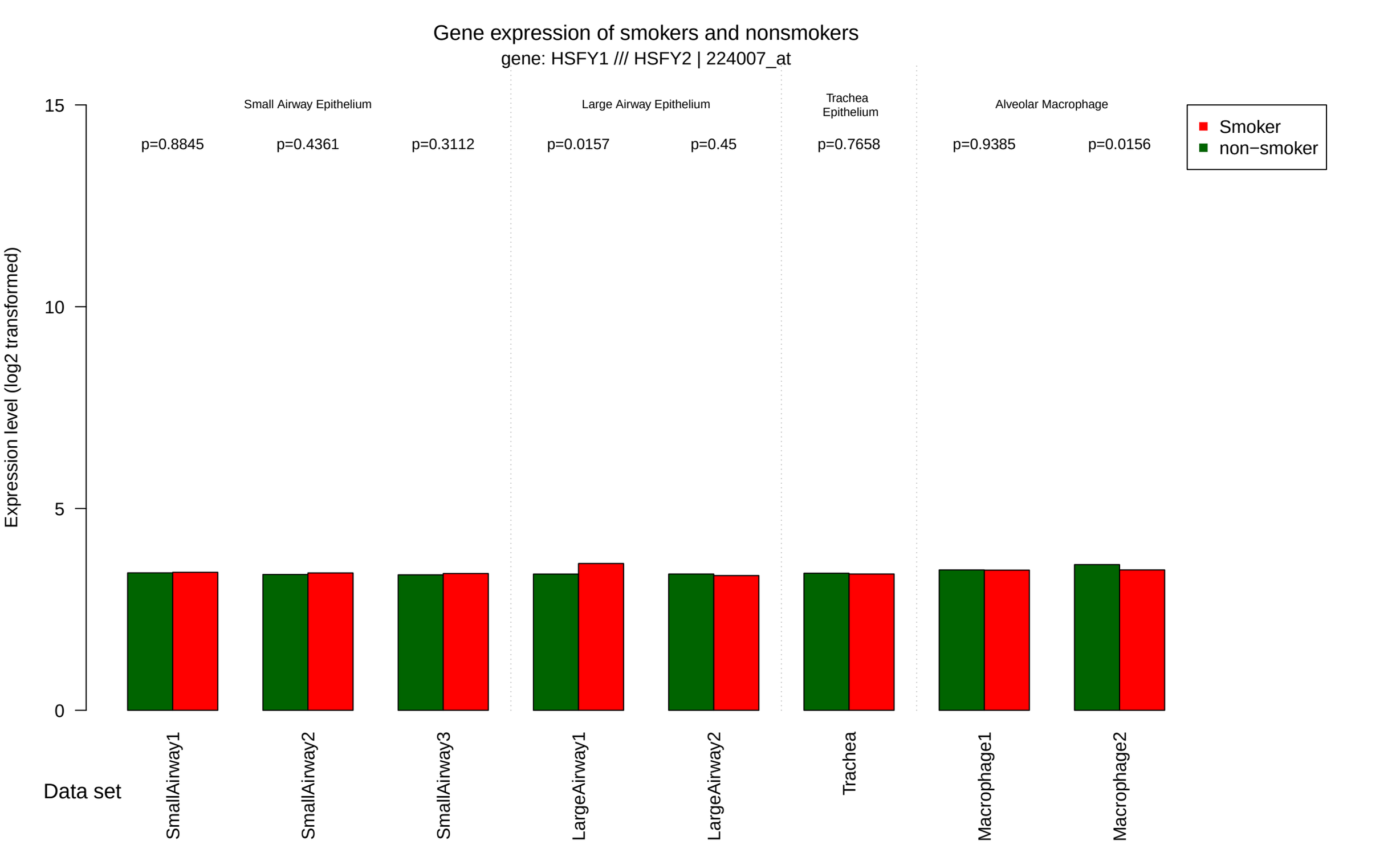
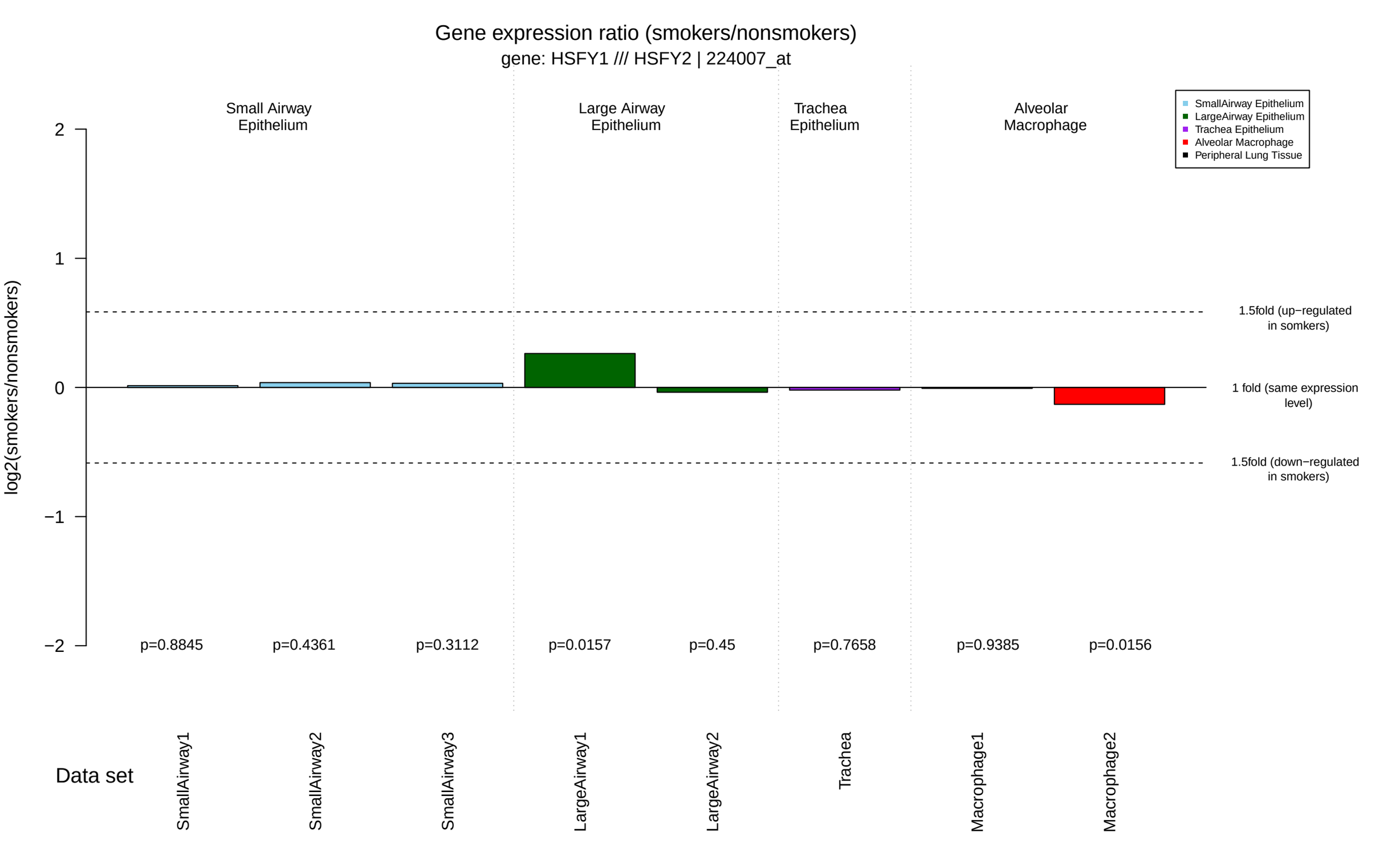
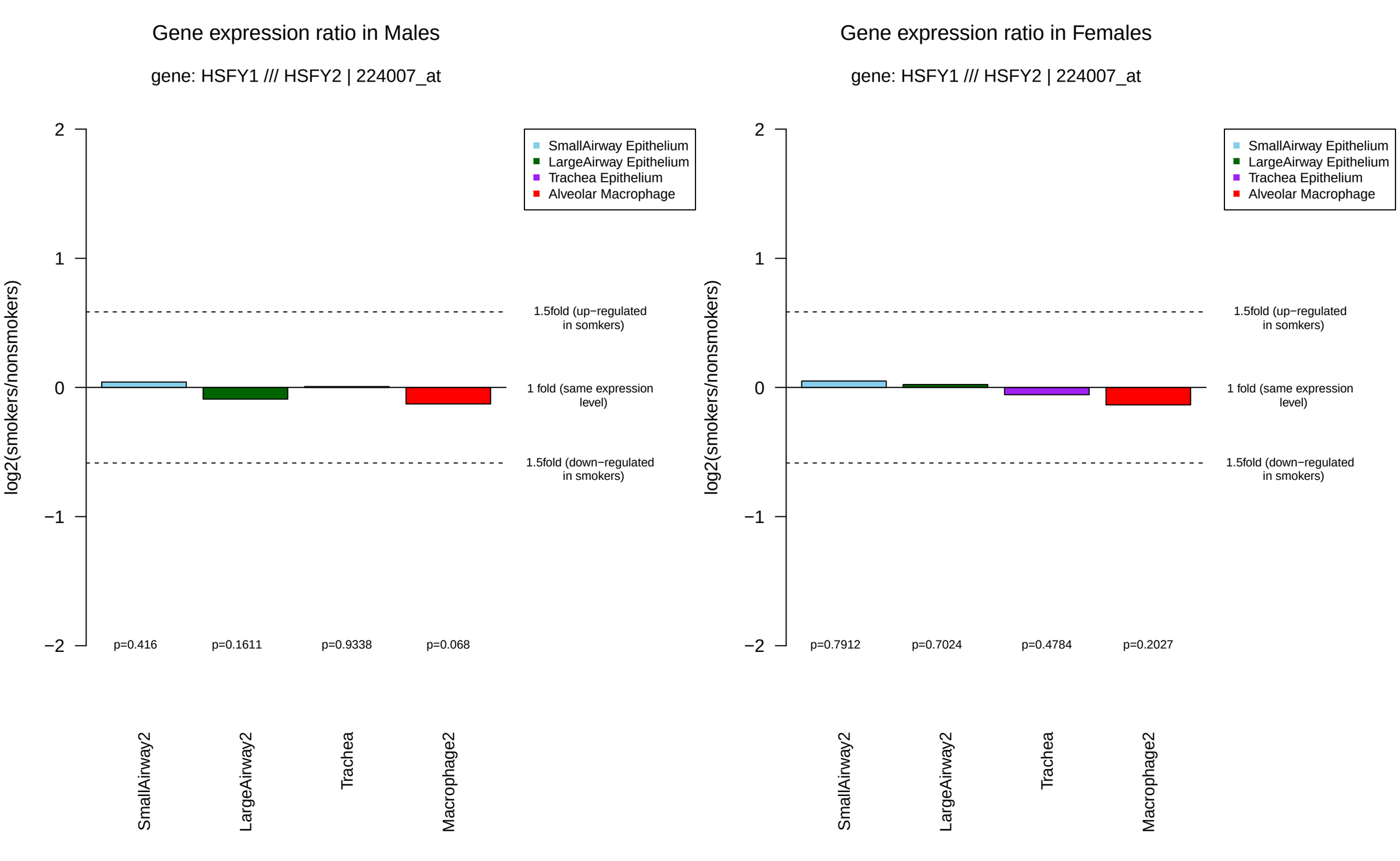
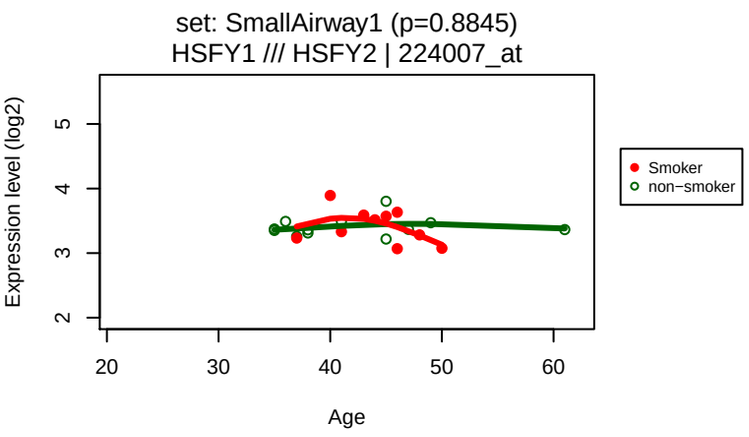
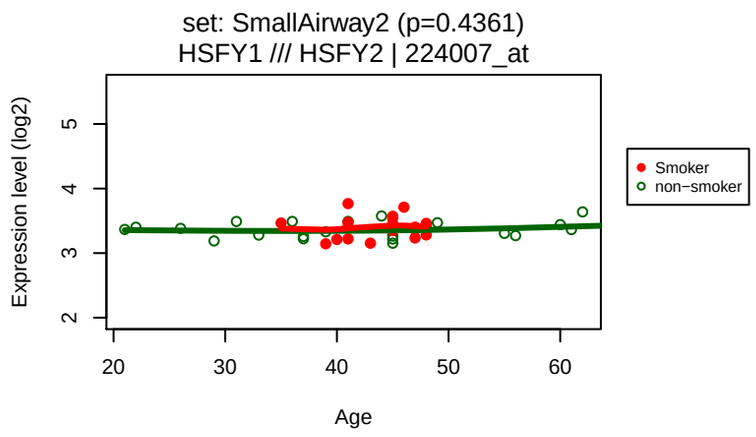
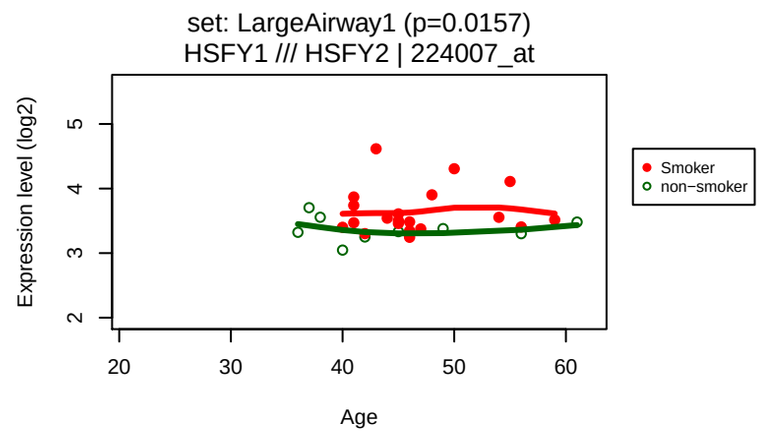

Gene expression of smokers and nonsmokers
gene: HSFY1 /// HSFY2 | 224007_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8845
p=0.4361
p=0.3112
p=0.0157
p=0.45
p=0.7658
p=0.9385
p=0.0156
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

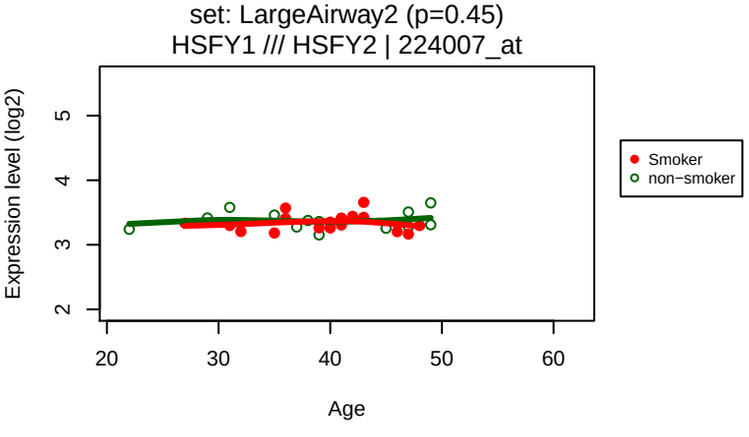

Gene expression ratio (smokers/nonsmokers)
gene: HSFY1 /// HSFY2 | 224007_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8845
p=0.4361
p=0.3112
p=0.0157
p=0.45
p=0.7658
p=0.9385
p=0.0156
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

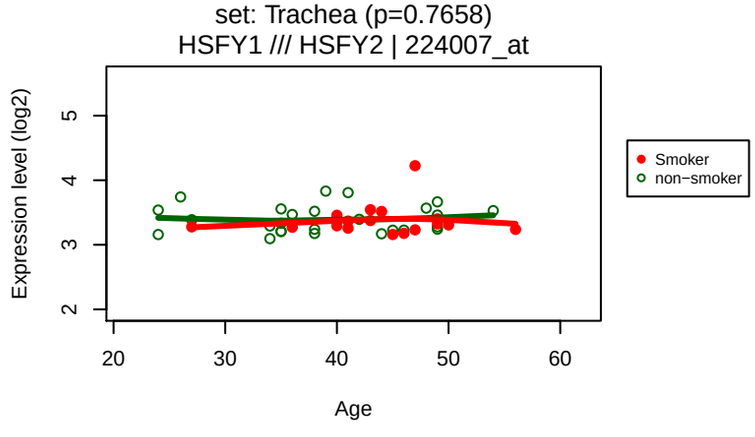

Gene expression ratio in Males
gene: HSFY1 /// HSFY2 | 224007_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.416
p=0.1611
p=0.9338
p=0.068
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSFY1 /// HSFY2 | 224007_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7912
p=0.7024
p=0.4784
p=0.2027
SmallAirway2
LargeAirway2
Trachea
Macrophage2

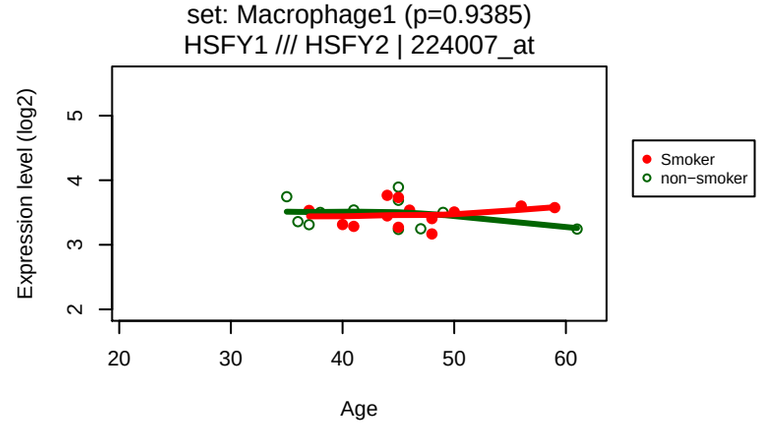

set: SmallAirway1 (p=0.8845)
HSFY1 /// HSFY2 | 224007_at
Expression level (log2)
Age
Smoker
non−smoker

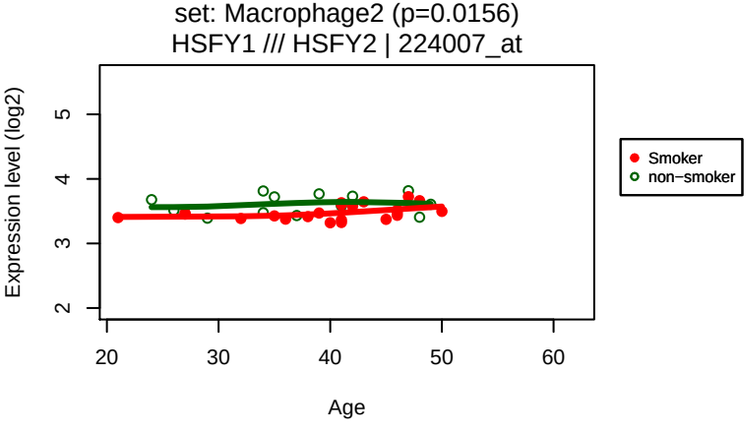

set: SmallAirway2 (p=0.4361)
HSFY1 /// HSFY2 | 224007_at
Expression level (log2)
Age
Smoker
non−smoker

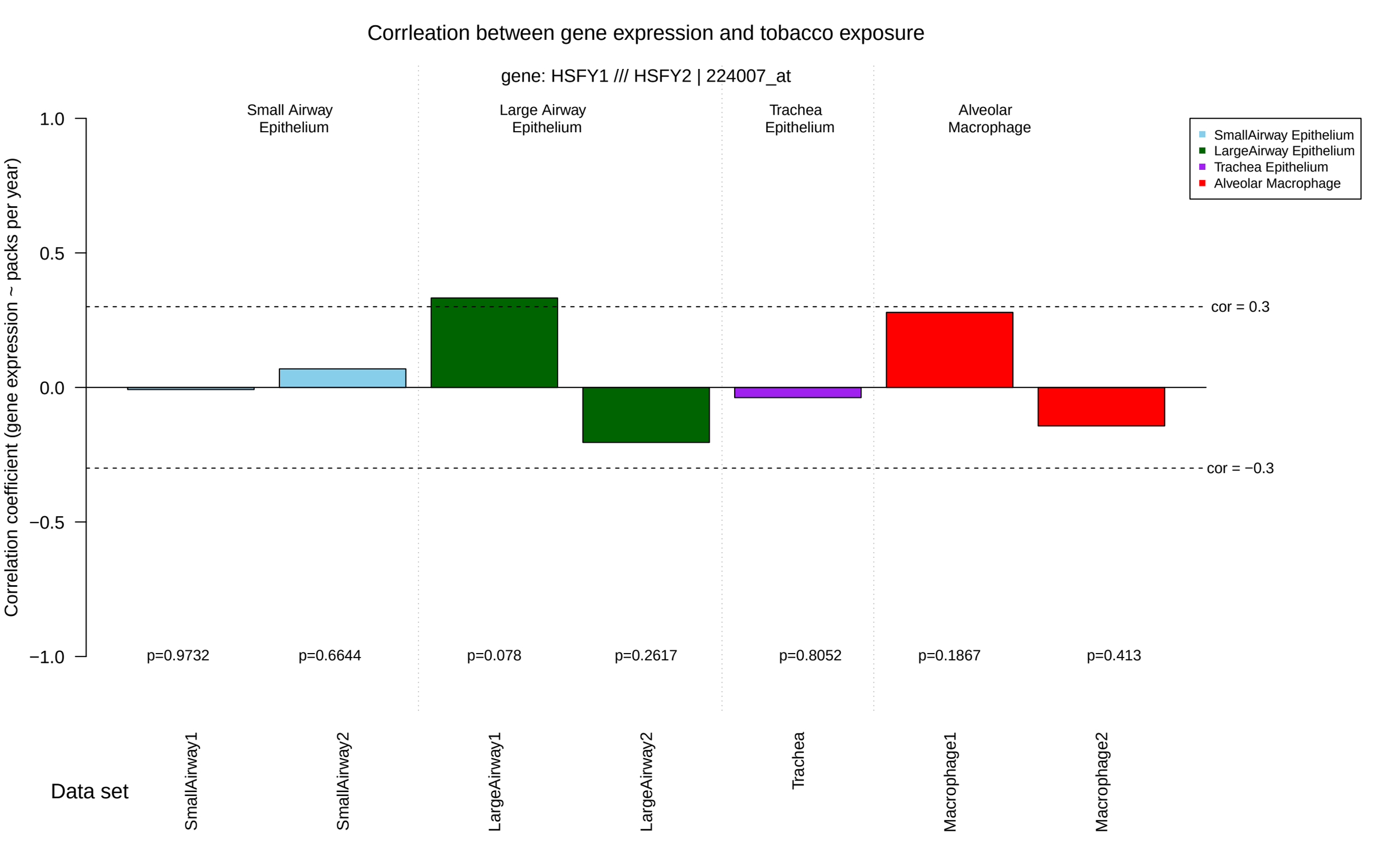

set: LargeAirway1 (p=0.0157)
HSFY1 /// HSFY2 | 224007_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.45)
HSFY1 /// HSFY2 | 224007_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7658)
HSFY1 /// HSFY2 | 224007_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9385)
HSFY1 /// HSFY2 | 224007_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0156)
HSFY1 /// HSFY2 | 224007_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSFY1 /// HSFY2 | 224007_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9732
p=0.6644
p=0.078
p=0.2617
p=0.8052
p=0.1867
p=0.413
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2